

BANARAS HINDU UNIVERSITY
B.A./B.Sc. (Hons) — Semester V Examination, 2022-23

Paper No. MTB-503: Programming in ‘C’

Time: 3 Hours

Full Marks: 50

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) What is the use of `printf()` and `scanf()` functions? [1]
- (b) Name four header files in the ‘C’ programming language. [1]
- (c) Determine the hierarchy of operations and evaluate the following expression, assuming `k` is a `float` variable: `k = 3/2*4+3/8`. [1]
- (d) What is the use of the function `toupper()`? Explain with an example code. [1]
- (e) If `a = 10`, `b = 12` and `c = 0`, find the values of: (i) `a != 6 && b > 5`; (ii) `!(a > 5 && c)`. [1]
- (f) What are the basic data types in ‘C’? [1]
- (g) What is the output of the following program?

```
tfamily
main() {
    int x = 101, y = 105;
    printf("%d", (x > y) ? x : y);
}
```

[1]

- (h) What is the difference between `++a` and `a++`? [1]
- (i) Describe the difference between `=` and `==` operators in ‘C’. [1]
- (j) Write the equivalent `while` loop code for:

```
tfamily
for (int i = 0; i < 5; i++)
    printf("%d", i);
```

[1]

Question 2.

- (a) Write a brief history of the 'C' programming language. [5]
 - (b) Write a 'C' program to find the Greatest Common Divisor (GCD) of two natural numbers. [5]
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Question 3.

- (a) What are the differences between local and global variables in 'C' with reference to declaration, visibility, and lifetime? [5]
 - (b) Write a general 'C' program to find the sum of the series $x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n-1}}{(2n-1)!}$ and print it on the screen. [5]
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Question 4.

- (a) What are entry-controlled and exit-controlled loops? Give examples. Using a **while** loop, write a 'C' program to evaluate $y = x^n$ where n is a non-negative integer. [5]
 - (b) Write a 'C' program to find the length of a string and count the number of vowels in it. [5]
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Question 5.

- (a) Write a 'C' program to print the multiplication table from 1×1 to 12×10 . [5]
 - (b) What is an array? Write the declaration, initialization and memory allocation of a one-dimensional array. [5]
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Question 6.

- (a) Write a 'C' program to compute and print the sum of all elements stored in an array using a pointer. [5]
 - (b) Write a 'C' program to: (i) print the word **MATHEMATICS**; (ii) count the occurrences and location of the letter 'A' in that word. [5]
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Question 7.

- (a) Write a 'C' program that uses functions and structures to perform: (i) reading a complex number; (ii) writing a complex number; (iii) addition of two complex numbers; (iv) multiplication of two complex numbers. [5]
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- (b) Write a 'C' program to read a series of three-digit integers from a file named **DATA**, then write all odd numbers to a file called **ODD** and all even numbers to a file called **EVEN**. [5]